The story of a radio image

How radiosynth turns a model of the corona
into a picture of the radio Sun (thermal free–free emission)

Soumitra Hazra

radiosynth 2.2.0 — *From MHD to MHz* — https://github.com/soumitrahazra/radiosynth

Imagine you point a radio telescope at the Sun and tune it to 80 MHz. You see a glowing disk, a little larger than the Sun you see with your eyes, at a temperature of about a million degrees. Where does that light come from, and how can a computer predict it from a simulation of the corona? This short note tells the story, one step at a time, with only the equations that are really needed.

1. We start at the telescope and walk backwards

The trick that makes the problem easy

Light travels from the Sun to the telescope. To make a picture we would have to follow every bit of light from every point of the corona and see which ones happen to reach us — hopeless. So we do the opposite: for every pixel of the image we start a *ray* at the telescope, send it backwards into the corona, and collect along the way all the light that would travel forward along the same path into that pixel.

The telescope sits at Earth, 215 solar radii away. Each pixel looks in a slightly different direction $\boldsymbol{v}$, and its ray starts where it first enters the model. One pixel, one ray, one number at the end: the brightness of that pixel.

In one sentence: one ray per pixel, traced from the observer into the model.

2. Where the light is born

Hot electrons passing by ions

The corona is a hot, thin gas of electrons and ions, at one to two million kelvin. Every time a fast electron flies past an ion, it is pulled and deflected, and it radiates a little burst of radio waves. This is *free–free emission*, or *bremsstrahlung* ("braking radiation"). Because the electrons move at random, like in any hot gas, the emission is *thermal*.

A hot gas that emits also absorbs. Two numbers describe it at each point:

- the **source function**, the brightness the gas would have if we could see only it, and nothing else. For a thermal gas at radio frequencies it is simply proportional to its temperature:

$$B_\nu = \frac{2\nu^2 k_\mathrm{B} T_e}{c^2}; \tag{1}$$

- the **absorption coefficient** κ_ν, which says how quickly the gas blocks light coming from behind it. Each electron needs an ion nearby, so it grows as the *square* of the density; slow electrons interact more, so it falls with temperature; and low frequencies are absorbed much more:

$$\kappa_\nu \simeq 0.018\, g\, \frac{n_e^2}{\nu^2\, T_e^{3/2}} \ \ \mathrm{cm}^{-1} \qquad (n_e \text{ in cm}^{-3},\ \nu \text{ in Hz},\ T_e \text{ in K}), \tag{2}$$

where $g \approx 10$ is the "Gaunt factor", a slowly varying number that collects the details of the collisions.

To get a feeling: in the low corona, $n_e = 10^9\,\mathrm{cm}^{-3}$ and $T_e = 1\,\mathrm{MK}$, a radio wave of 80 MHz is weakened by a factor e after only about 360 km ($\kappa \approx 2.8 \times 10^{-6}\,\mathrm{m}^{-1}$), while at 1 GHz it can travel about 56 000 km. That is why the low-frequency Sun is opaque high in the corona, and the high-frequency Sun is transparent.

In one sentence: every point of the corona glows at its own temperature (B_ν) and blocks light behind it at a rate $\kappa_\nu \propto n_e^2/\nu^2$.

3. The corona is a lens

Why the rays do not go straight

A plasma slows down radio waves of low frequency. Its refractive index is

$$\mu = \sqrt{1 - \frac{n_e}{n_{\mathrm{cr}}}}, \qquad n_{\mathrm{cr}} \approx 1.24 \times 10^8 \left(\frac{\nu}{100\,\mathrm{MHz}}\right)^2 \ \mathrm{cm}^{-3}. \tag{3}$$

n_{cr} is the *critical density*. Where the density approaches it, μ drops towards zero; beyond it ($n_e > n_{\mathrm{cr}}$) the wave cannot exist at all. The corona is densest near the Sun, so it acts like a lens that pushes the rays *away* from the dense plasma. The direction of the ray turns according to

$$\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}s} = \frac{\nabla\mu}{\mu} - \boldsymbol{v}\left(\boldsymbol{v}\cdot\frac{\nabla\mu}{\mu}\right), \tag{4}$$

which simply says: *bend towards where μ is larger*, i.e. towards thinner plasma. A ray that heads for the Sun slows down, curves, and is reflected before it reaches the critical surface — like a ball rolling up a hill and coming back down (Fig. 1). At 80 MHz the critical surface of a typical corona lies at about $1.3\,R_\odot$, so the rays never get closer to the Sun than that.

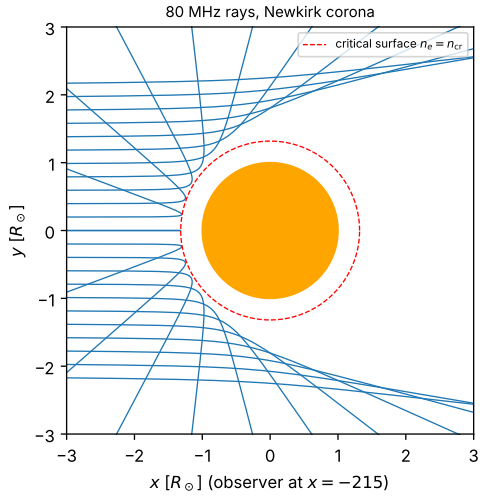


Figure 1: Rays at 80 MHz, computed by radiosynth. They come from the telescope on the left, curve away from the dense plasma and turn around above the critical surface (red dashed).

In one sentence: the denser the plasma, the more the rays bend; they turn back before the critical density $n_{\mathrm{cr}} \propto \nu^2$.

4. Walking the ray, one step at a time

What the computer actually does

The computer cannot follow a smooth curve; it takes small steps of length Δs. At each step, for every ray at once, it does the same few things:

1. **Look half a step ahead.** Go to the middle of the next step, $\boldsymbol{r}_{1/2} = \boldsymbol{r} + \boldsymbol{v}\,\Delta s/2$, and read the density, its gradient and the temperature of the model there.
2. **Check that the ray can continue.** Stop if it has left the model, hit the photosphere, or run into plasma denser than $n_{\rm cr}$.
3. **Make sure the step is small enough.** If the ray would bend too much in one step (more than about 1/50 of a radian), or if it is about to reach the critical surface, shrink the step and try again.
4. **Turn and move.** Rotate the direction by the amount the lens of Chapter 3 asks for, and move to the end of the step. Right at the turning point, where the path is a small parabola, jump directly to the other side of the parabola. If a ray overshoots anyway, step back and, as a last resort, reflect it like light on a mirror.
5. **Collect the light** of this step (Chapter 5).
6. **Grow the step again** towards the size of the model's cells, so that the ray does not crawl where nothing happens.

This recipe is the one invented for the BATS-R-US code by Benkevitch et al. (2010) and used by Moschou et al. (2018); radiosynth follows it, with the rotation done by the Boris method, which keeps the direction exactly of unit length.

In one sentence: small, careful steps: look ahead, check, shrink if needed, bend, move, collect.

5. Collecting the light

Glowing, but behind a veil

Each step is a small piece of glowing gas of length $\Delta\ell$. Its *optical depth*

$$\Delta\tau = \kappa_\nu\,\Delta\ell \tag{5}$$

says how opaque it is: $\Delta\tau \ll 1$ is transparent, $\Delta\tau \gg 1$ is a wall.

The piece emits $B_\nu(1 - e^{-\Delta\tau})$: almost nothing if it is transparent, its full brightness B_ν if it is a wall. But we see it through all the gas that lies between it and the telescope, which has already piled up an optical depth τ and lets only a fraction $e^{-\tau}$ through. So the ray adds

$$\boxed{I_\nu \;\leftarrow\; I_\nu + B_\nu\left(1 - e^{-\Delta\tau}\right)e^{-\tau}, \qquad \tau \;\leftarrow\; \tau + \Delta\tau} \tag{6}$$

and moves on. Two things follow:

- Once τ reaches a few, the veil in front is so thick that nothing deeper can be seen; the code stops the ray when $\tau > 30$.
- If the gas is opaque and has the same temperature everywhere, the sum gives exactly $I_\nu = B_\nu$: the ray "sees" the temperature of the gas, no more.

There is also a simpler, *optically thin* version, $I_\nu = \sum \kappa_\nu B_\nu \, \Delta\ell$, which just adds everything and forgets the veil. It is what BATS-R-US does, and it is fine where the corona is transparent, but in the dense low corona it adds too much and can give brightness temperatures higher than the gas itself (Fig. 2, right). The full version is used with `transfer="full"`.

In one sentence: each step adds its own glow, dimmed by everything in front of it.

6. From numbers to a picture

Brightness temperature and flux

Radio astronomers like to express brightness as a temperature: the temperature an opaque gas would need to be this bright,

$$T_b = \frac{c^2}{2 k_\mathrm{B} \nu^2} \, I_\nu . \tag{7}$$

Where the corona is opaque, T_b is simply the temperature of the layer where the ray became opaque; where it is transparent, T_b is smaller. Adding up all the pixels, each covering a small solid angle $\Delta\Omega = \Delta X \Delta Y / D^2$ on the sky, gives the total flux the telescope measures:

$$F_\nu = \sum_{\text{pixels}} I_\nu \, \Delta\Omega \qquad (\text{in sfu} = 10^{-22}\,\mathrm{W\,m^{-2}\,Hz^{-1}}). \tag{8}$$

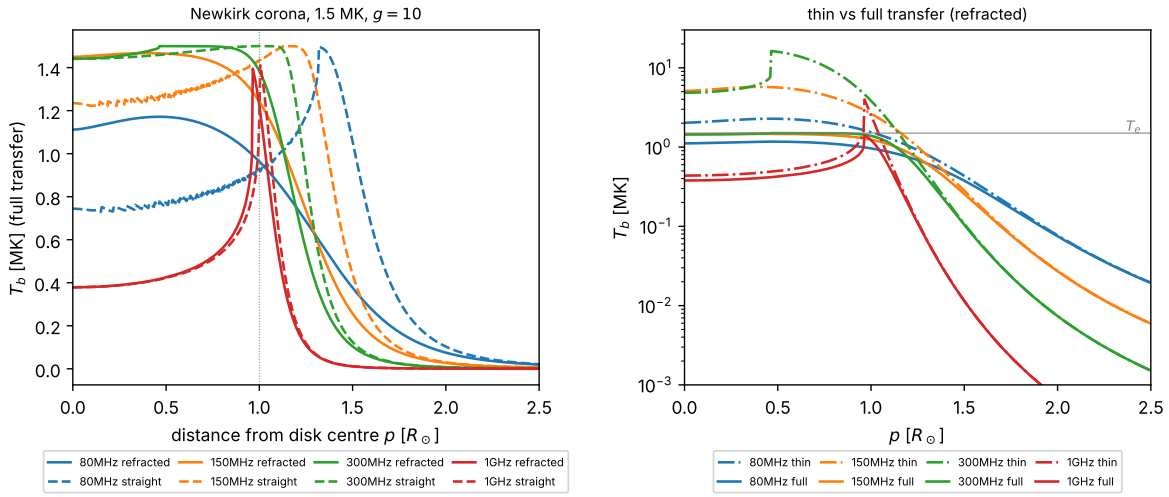


Figure 2: The result for a simple model corona at 1.5 MK: brightness temperature from the centre of the disk ($p = 0$) outwards. Left: with refraction (solid) and with straight rays (dashed). Right: full transfer (solid) and the thin sum (dash-dotted), which overshoots T_e.

For the model corona of Fig. 2, the code finds at 80 MHz a disk of about 1 MK that extends beyond the visible Sun, and a total flux of about 3 sfu — close to what radio telescopes see from the quiet Sun. At 1 GHz the corona has become nearly transparent: the disk is the size of the optical Sun, fainter in the centre and brighter at the limb, where the line of sight through the corona is longest.

In one sentence: T_b tells us the temperature of the layer we see; the sum over pixels gives the flux.

7. What the lens changes

Refraction on or off

radiosynth can switch refraction off (`no_refraction=True`) and compare the two images (`rs.compare_refraction`). Fig. 2 (left) shows what happens at low frequency. With refraction the radio Sun is *smaller*, because rays that pass near the limb are bent away and see less dense plasma (3.1 instead of 4.1 sfu at 80 MHz). Its centre, however, is *brighter*: a straight ray stops at the critical surface before the gas has become opaque, while a bent ray turns above it and spends a long time in the dense plasma, so it collects more light. At 1 GHz the density is far below $n_{\rm cr}$, the lens is gone, and both images agree.

In one sentence: refraction shrinks the low-frequency radio Sun and changes its brightness; at high frequency it does not matter.

8. The whole story in five lines

1. For each pixel, start a ray at the telescope and send it into the model.
2. Bend it with the refractive index $\mu = \sqrt{1 - n_e/n_{\rm cr}}$, in small, controlled steps.
3. At each step, the gas glows at $B_\nu \propto T_e$ and absorbs at $\kappa_\nu \propto n_e^2 \nu^{-2} T_e^{-3/2}$.
4. Add the glow of each step, dimmed by the gas in front: $I \leftarrow I + B(1 - e^{-\Delta\tau})e^{-\tau}$.
5. Convert to T_b and sum the pixels to get the flux.

In radiosynth all of this is one line:

```
img = rs.make_radio_image(grid, "80MHz, 150MHz, 1GHz", transfer="full")
```

and the details, with every equation and the code that implements it, are in the companion note *Thermal free–free (bremsstrahlung) emission in radiosynth*.

Further reading. Benkevitch et al. 2010, ApJ 725, 447 (the ray-tracing algorithm); Moschou et al. 2018, ApJ 867, 51 (radio images from MHD models); Dulk 1985, ARA&A 23, 169 (radio emission from the Sun and stars).